

$$x_1 = 3 \quad x_2 = 3$$

weighted sum.

$$z = w_1(x_1) + w_2(x_2) + b$$

$$= 2(3) + 1.5(3) - 15$$

$$= -6.5$$

Sigmoid = $\frac{1}{1+e^{-z}}$ lets say the answer is

weights will decide how good the model

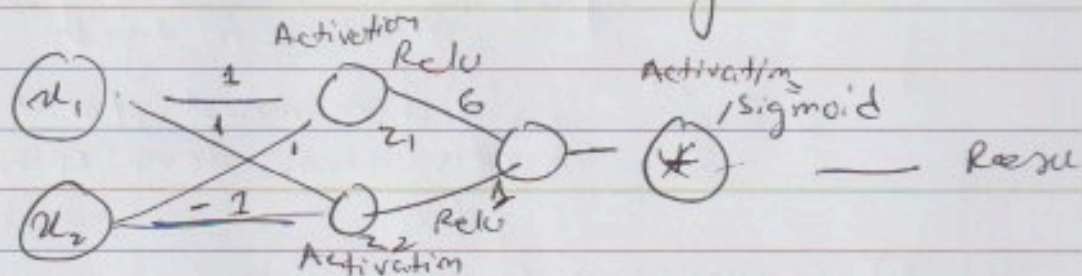
Bias : Some adjustment the model makes

Activation function

Sigmoid

Relu (0,t) = max(0,t) max of 0 to t

if we add hidden layer.



$$z_1 = 1(x_1) + 1(x_2) - 5 \quad \text{out} = \text{Relu}(z_1)$$

$$z_2 = 1(x_1) - 1(x_2) + 0 \quad \text{out} = \text{Relu}(z_2)$$

$$z_1 = 1(6) + 1(5) - 5 = 6 \quad (0, 6)$$

Since Relu take max(0,6), Relu is (0,t)

$$z_2 = 1(6) - 1(5) - 0 = 1 \quad \text{Relu}(0, 1) = 1$$

$$z_{\text{out}} = 6(1) + 1(1) - 6 = 1 = \frac{1}{1+e^{-z}} = 0.73 \text{ or something}$$